Security frameworks

NIST cybersecurity framework

Identify	Protect	Detect	Respond	Recover
• Threat intelligence • Vulnerability management	• Zero-trust architecture • Secrets Management • Identity and Access Management (IAM) • Immutable Infrastructure as Code (IaC) • Secure Software Supply Chain	• Security observability • Detection engineering • Security monitoring	• Automated security testing • Continuous compliance frameworks	• Immutable Infrastructure as Code (IaC) • Continuous compliance frameworks

The **National Institute of Standards** and **Technology (NIST) Cybersecurity Framework** helps organizations better understand, manage, and reduce their cybersecurity risk and protect their networks and data.

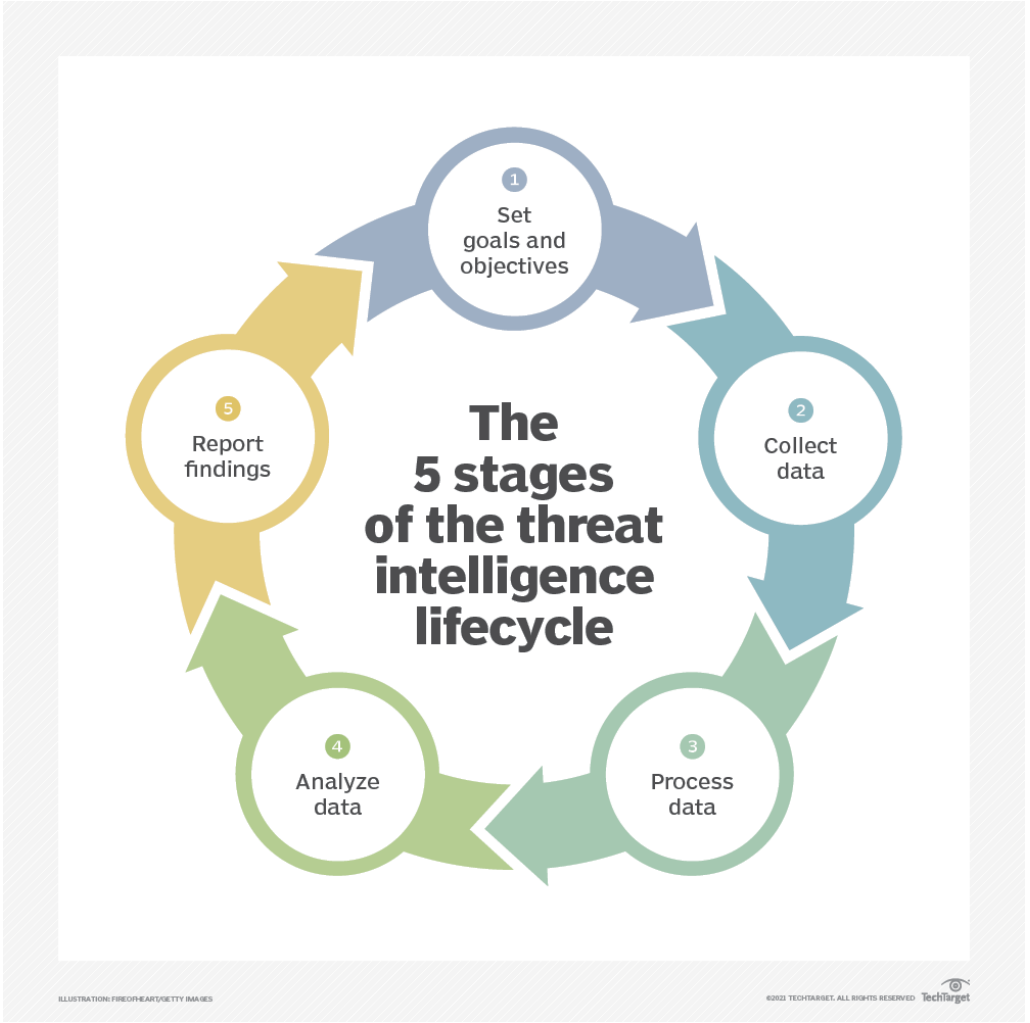


Source: https://genai.works/courses/nist-cybersecurity-and-risk-management-frameworks-specialization

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Threat Intelligence

Shared threat intelligence ensures both DevSecOps teams are aware of emerging risks and can act consistently on the same data. DevSecOps integrates threat intelligence into CI/CD pipelines.



Source: https://www.techtarget.com/whatis/definition/threat-intelligence-cyber-threat-intelligence

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Vulnerability management

Vulnerability management aligns pre-deployment scanning with runtime vulnerability detection, ensuring continuous risk reduction. DevSecOps identifies vulnerabilities early.

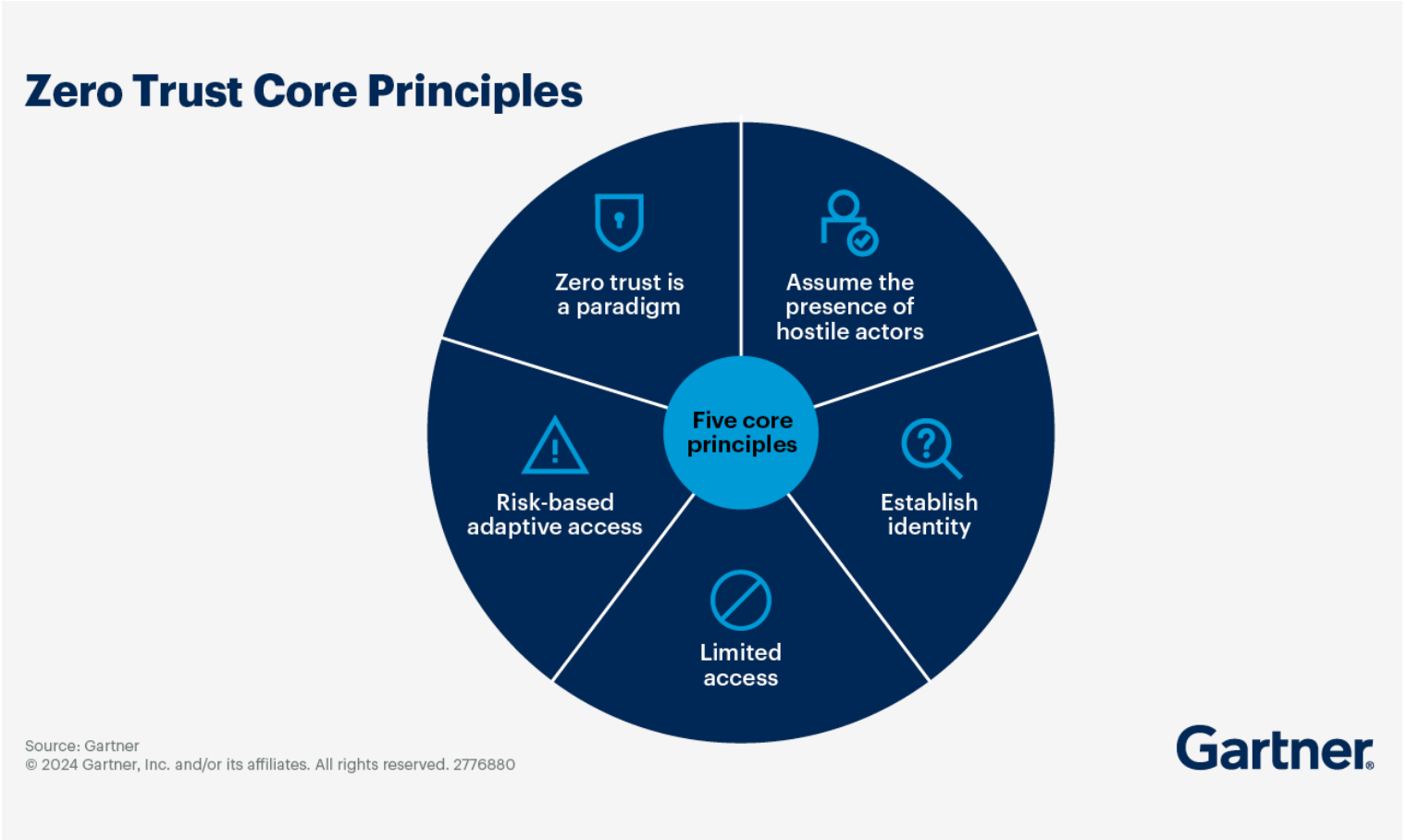


Source: https://www.linkedin.com/pulse/vulnerability-management-lifecycle-yotam-perkal/

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Zero-trust architecture

Zero-trust enforces continuous verification of users, devices, and applications across both development and production environments. This eliminates trust gaps and ensures consistent security enforcement. DevSecOps benefits from secure pipelines.



Source: https://www.gartner.com/en/cybersecurity/topics/zero-trust-architecture

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Secrets management

Secure secrets handling aligns both DevSecOps teams by providing consistent management for credentials, keys, and tokens across CI/CD pipelines and runtime operations. DevSecOps focuses on secure development.

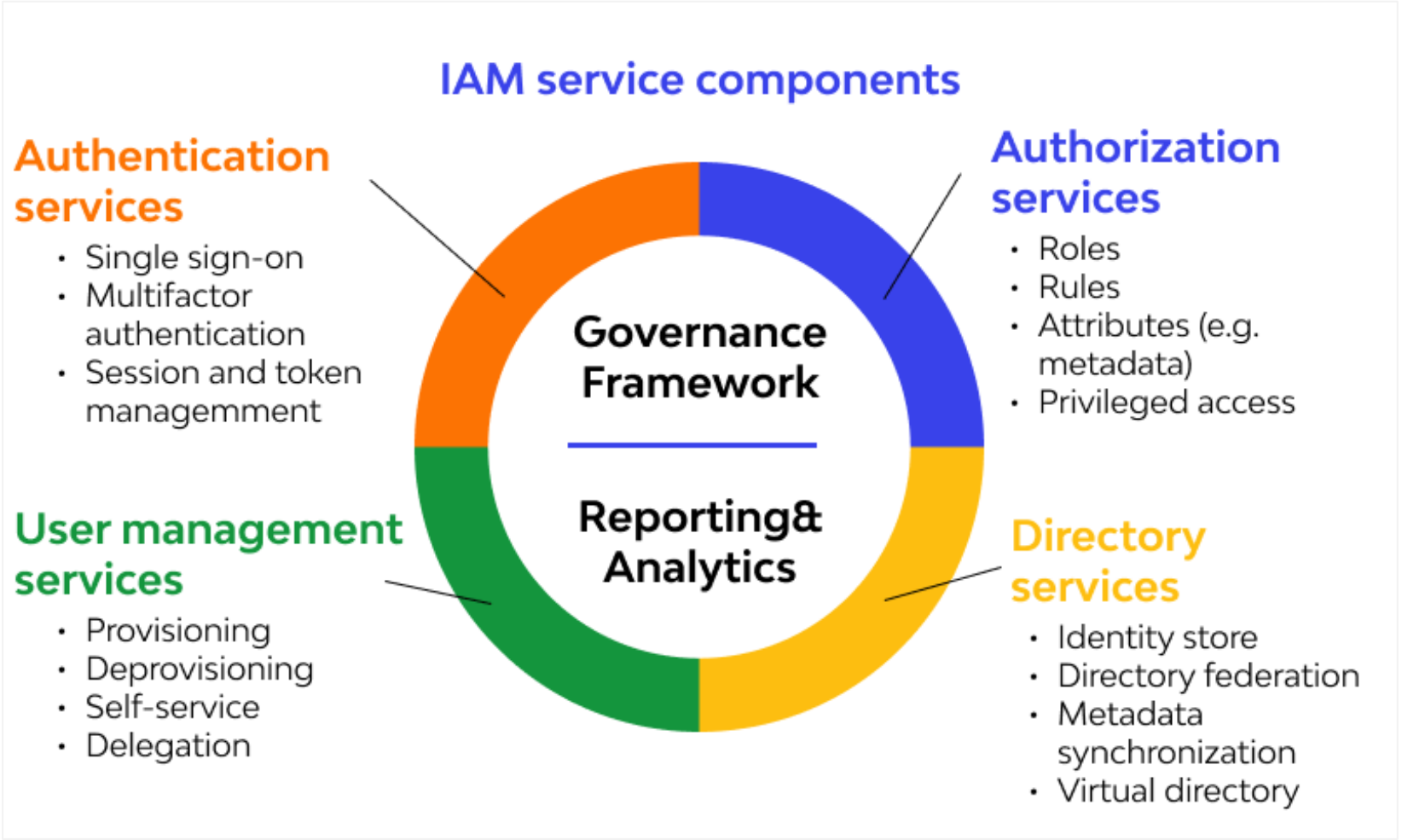


Source: https://www.syteca.com/en/blog/secrets-management

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Identity and Access Management (IAM)

IAM enforces consistent access control policies, reducing identity-based threats. DevSecOps secures developer access.

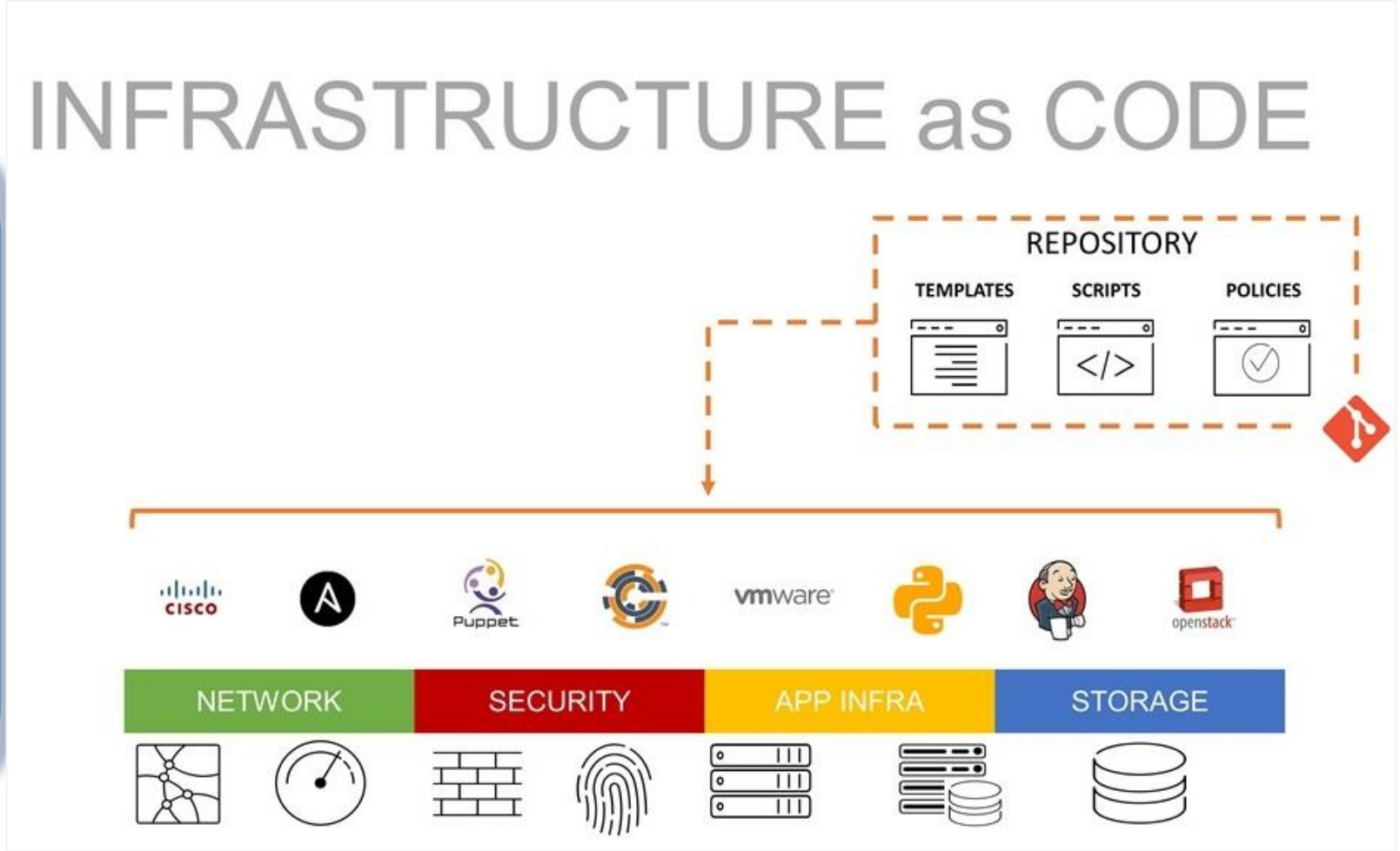


Source: https://www.csub.edu/its/security/account-information/identity-and-access-management.shtml

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Immutable Infrastructure as Code (IaC)

Immutable IaC ensures that both DevSecOps teams work with identical, secure infrastructure definitions, eliminating drift between development and production. DevSecOps uses IaC to define secure build and test environments.

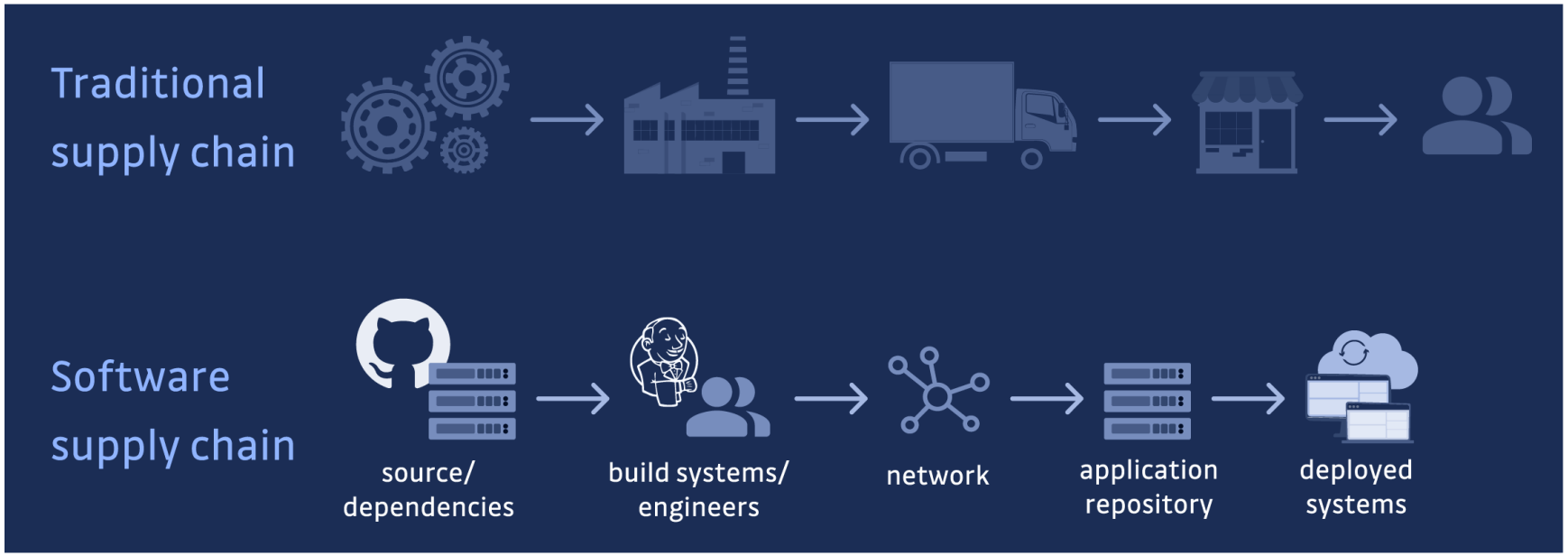


Source: https://www.f5.com/glossary/infrastructure-as-code-iac

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Secure software supply chain

Secure supply chain practices ensure consistency in securing dependencies and artifacts across both development. DevSecOps integrates security into build pipelines.

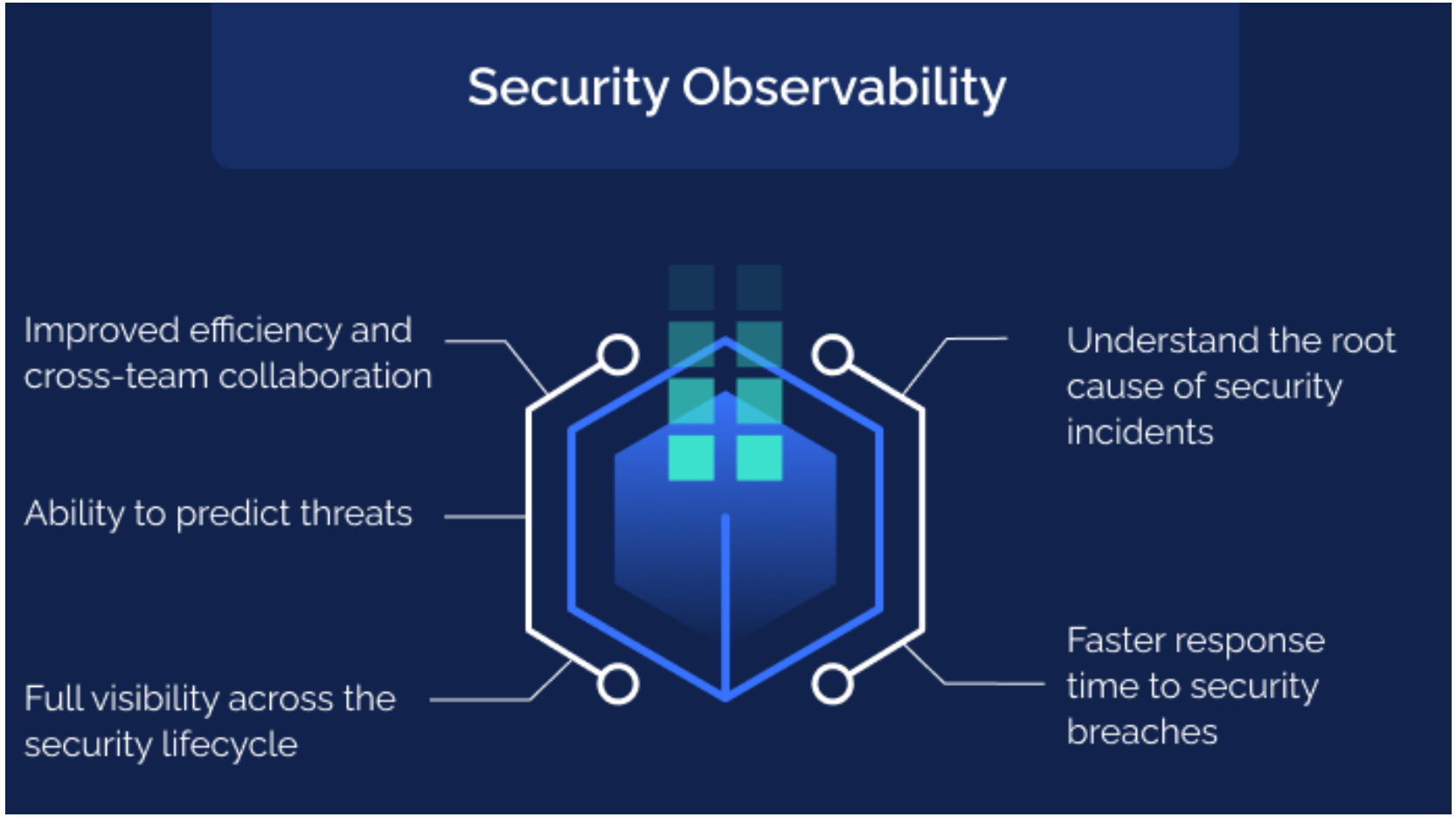


Source: https://www.f5.com/glossary/infrastructure-as-code-iac

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Security observability

Observability bridges visibility gaps by correlating logs, metrics, and traces from pre-deployment environments (DevSecOps). Teams gain real-time insights into potential vulnerabilities and operational anomalies.

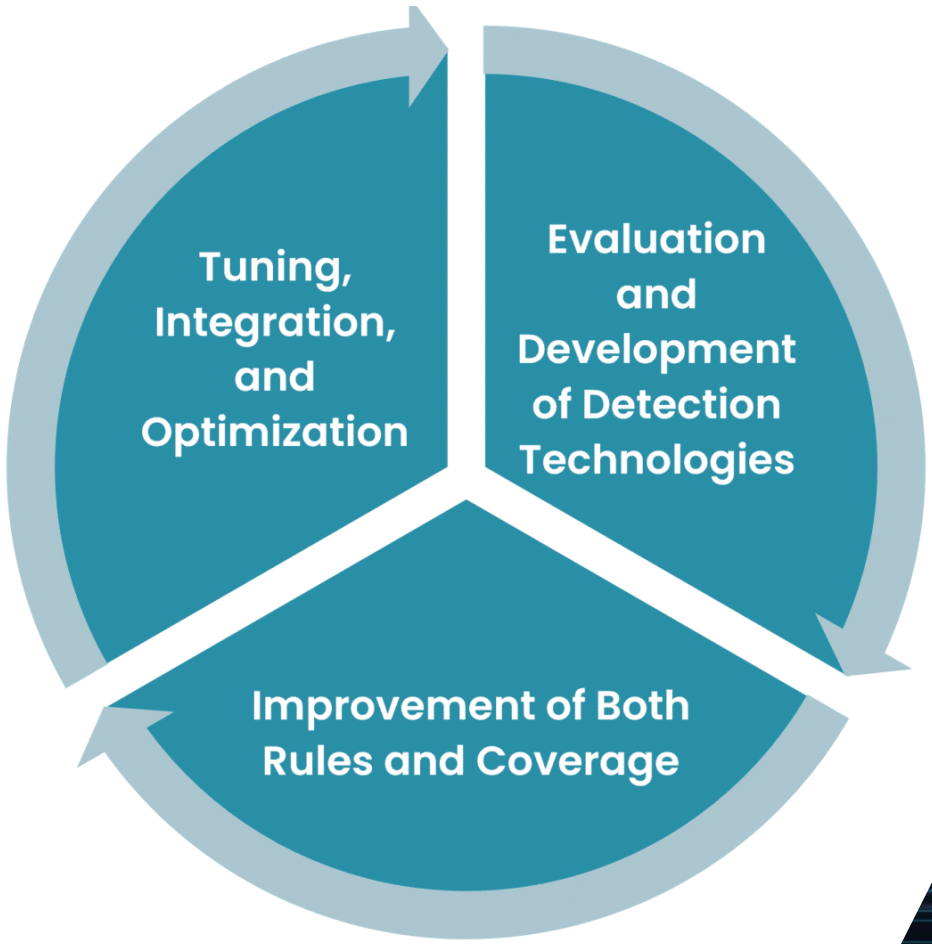


Source: https://www.eteam.io/blog/combine-observability-driven-development-and-security

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Detection engineering

Detection engineering ensures consistent threat detection rules that are applied during development, testing, and production. DevSecOps uses detection rules to build secure software.



Source: https://www.truesec.com/security/detection-engineering-a-strategic-imperative



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Security monitoring

Security monitoring combines pre-deployment testing (DevSecOps) with runtime monitoring to provide end-to-end visibility. Unified monitoring ensures no gaps between testing and production.



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Automated security testing



Automated testing ensures vulnerabilities are identified consistently during development (DevSecOps) and in staging/production.

Source: https://www.techmagic.co/blog/automated-security-testing

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DevSecOps compliance framework

DevSecOps compliance ensures adherence to security policies in both CI/CD pipelines. DevSecOps validates compliance during development.



Source: https://www.opswat.com/blog/cybersecurity-compliance

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